

Middleware Architectures 1

Lecture 5: Cloud Native and Kubernetes

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Overview

- Cloud Native
- Kubernetes

Overview

- The Cloud Native Computing Foundation (CNCF)
 - *Motto: Building sustainable ecosystems for cloud native software*
 - *CNCF is part of the nonprofit Linux Foundation*
- Cloud Native = scalable apps running in modern cloud environments
 - *containers, service meshes, microservices*
 - *Apps must be usually re-built from scratch or refactored*
 - *Benefits:*
 - *loosely coupled systems that are resilient, manageable, and observable*
 - *automation allowing for predictable and frequent changes with minimal effort*
 - *Trail Map*
 - *provides an overview for enterprises starting their cloud native journey*
- Lift and Shift
 - *Cloud transition program in organizations*
 - *Move app from on-premise to the cloud*
 - *Benefits*
 - *Infrastructure cost cutting (OPEX vs. CAPEX)*
 - *Improved operations (scaling up/down if possible can be faster)*

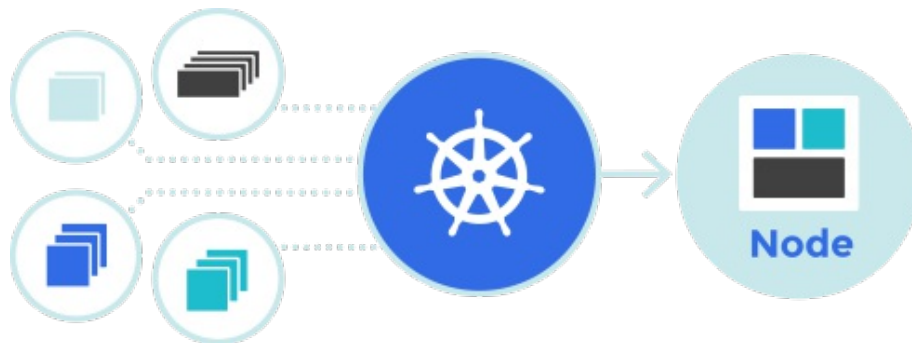
CNCF Trail Map

Overview

- Cloud Native
- Kubernetes
 - *Basic Concepts*
 - *Core Concepts and Architecture*
 - *Workloads*
 - *Services*
 - *Beyond the Basics*

Overview

- In your architecture...
 - Containers are atomic pieces of application architecture
 - Containers can be linked (e.g. web server, DB)
 - Containers access shared resources (e.g. disk volumes)
- Kubernetes
 - Automation of deployments, scaling, management of containerized applications across number of nodes
 - Based on Borg, a parent project from Google

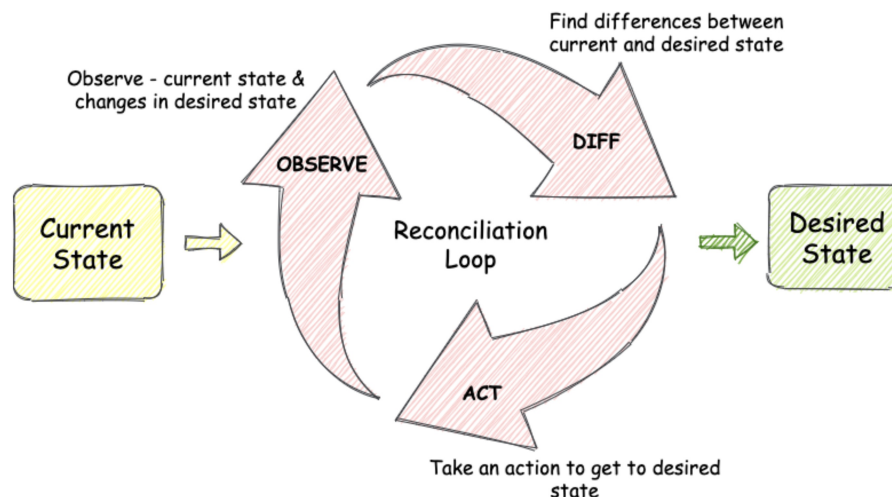


Key Design Principles

- Kubernetes provides abstractions that separate application deployment from the underlying infrastructure details
- Application workloads and infrastructure decoupling
 - **Compute:** *Define what to run without specifying where it runs*
 - **Storage:** *Applications request storage independent of storage backend*
 - **Networking:** *Stable access to applications regardless of IPs or location*
- Benefits
 - *Portability across on-prem and cloud environments*
 - *Scalability and resilience through dynamic scheduling*
 - *Consistency and standardization of deployment model*
 - *Reduced vendor lock-in thanks to open standards*

Desired State and Reconciliation

- Kubernetes operates on a **desired state** model
 - Users define the state they want through object specifications (YAML)
 - Example: “there should be 3 replicas of this application”
- Actual State vs. Desired State
 - Kubernetes constantly monitors the cluster
 - If the actual state drifts from the desired state, it takes action to fix it
- Reconciliation Loop
 - Controllers continuously compare desired vs. actual state
 - Automatically performs actions such as restarting, rescheduling, or scaling Pods



Features

- Automatic binpacking
 - *Automatically places containers onto nodes based on their resource requirements and other constraints.*
- Horizontal scaling
 - *Scales your application up and down with a simple command, with a UI, or automatically based on CPU usage.*
- Automated rollouts and rollbacks
 - *Progressive rollout out of changes to application/configuration, monitoring application health and rollback when something goes wrong.*
- Storage orchestration
 - *Automatically mounts the storage system (local or in the cloud)*
- Self-healing
 - *Restarts containers that fail, replaces and reschedules containers when nodes die, kills containers that don't respond to user-defined health checks.*
- Service discovery and load balancing
 - *Gives containers their own IP addresses and a single DNS name for a set of containers, and can load-balance across them.*

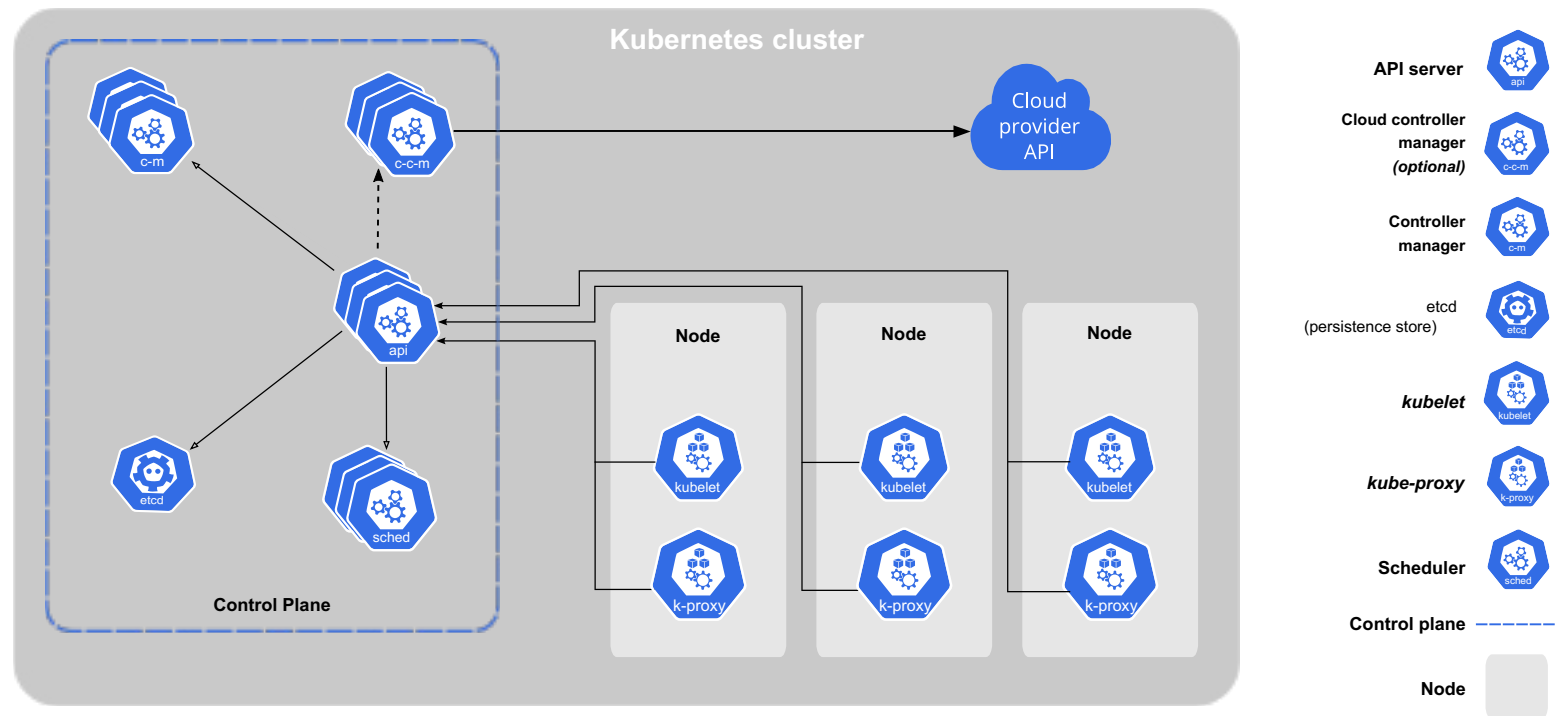
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Core Building Blocks

- **Cluster**
 - *A set of worker nodes and a control plane*
 - *Runs and manages containerized applications*
- **Node**
 - *A worker machine in Kubernetes (VM or physical)*
 - *Runs Pods scheduled by the control plane*
- **Control Plane**
 - *Manages the overall state of the cluster*
 - *Schedules workloads and responds to cluster events*
- **Pod**
 - *The smallest deployable unit in Kubernetes*
 - *One or more tightly-coupled containers*
 - *Containers share networking and storage within a Pod*

Architecture



Control Plane Components (Part 1)

- Global decisions about the cluster
 - *Schedulling*
 - *Detecting and responding to cluster events, starting up new pods*
- kube-apiserver
 - *exposes the Kubernetes API*
 - *The API server is the front end for the Kubernetes control plane.*
- etcd
 - *highly-available key value store used to store all cluster data*
- kube-scheduler
 - *watches for newly created Pods with no assigned node*
 - *selects a node for Pods to run on.*
 - *Decision factors: resource requirements, hardware/software/policy constraints, affinity and anti-affinity specifications*

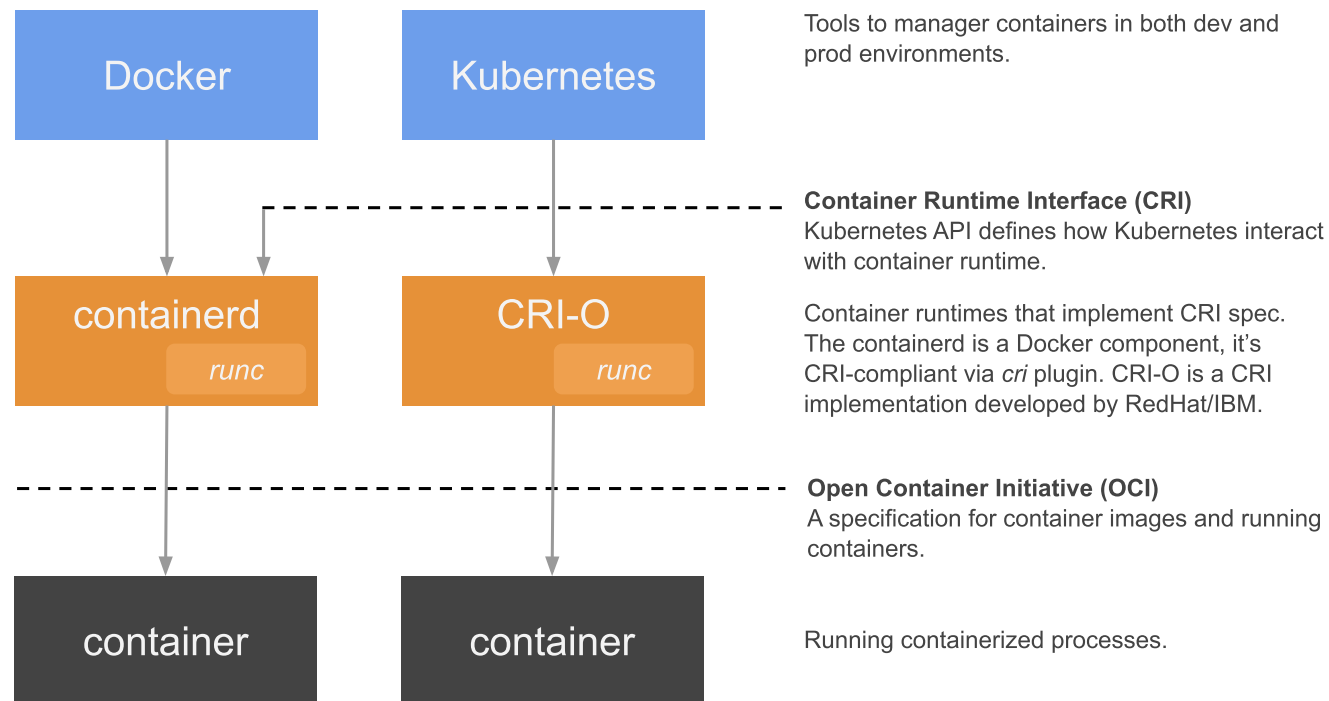
Control Plane Components (Part 2)

- kube-controller-manager
 - *runs controller to ensure the desired state of cluster objects*
 - **Node controller**
 - *noticing and responding when nodes go down*
 - **Job controller**
 - *creates Pods to run one-off tasks to completion.*
 - **Endpoints controller**
 - *Populates the Endpoints object (that is, joins Services, Pods).*
- cloud-controller-manager
 - *Integration with cloud services (when the cluster is running in a cloud)*
 - **Node controller**
 - *checks if a node has been deleted in the cloud after it stops responding*
 - **Route controller**
 - *For setting up routes in the underlying cloud infrastructure*
 - **Service controller**

Node

- Kubernetes runtime environment
 - *Run on every node*
 - *Maintaining running pods*
- kubelet
 - *An agent that runs on each node in the cluster*
 - *It makes sure that containers are running in a Pod.*
- kube-proxy
 - *maintains network rules on nodes*
 - *network rules allow network communication to Pods from inside or outside of the cluster*
 - *uses the operating system packet filtering layer or forwards the traffic itself.*
- Container runtime
 - *Responsible for running containers*
 - *Kubernetes supports several container runtimes (containerd, CRI-O)*
 - *Any implementation of the Kubernetes CRI (Container Runtime*

Container Stack



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Namespaces

- Logical grouping of cluster resources
 - *Allow you to organize and separate objects within a Kubernetes cluster*
 - *Useful when multiple teams, environments, or projects share the same cluster*
- Rationale
 - *Provide isolation and boundaries between workloads*
 - *Prevent name collisions*
 - *Objects can have the same name if in different namespaces*
 - *Enable resource limits and access control per namespace*
- Usage
 - *Common namespaces: default, kube-system, kube-public, kube-node-lease*
 - *Create separate namespaces for e.g. dev, test, prod*
 - *Commands run in a namespace unless another is specified*

Pod

- Pod
 - *A group of one or more tightly-coupled containers.*
 - *Containers share storage and network resources.*
 - *A Pod runs a single instance of a given application*
 - *Pod's containers are always co-located and co-scheduled*
 - *Pod's containers run in a shared context, i.e. in a set of Linux namespaces*
- Pods are created using workload resources
 - *You do not create them directly*
- Pods in a Kubernetes cluster are used in two main ways
 - *Run a single container, the most common Kubernetes use case*
 - *Run multiple containers that need to work together*

Workloads

- An application running on Kubernetes
- Workloads run in a set of Pods
- Pre-defined workload resources to manage lifecycle of Pods
 - **Deployment and ReplicaSet**
 - *managing a stateless application workload*
 - *any Pod in the Deployment is interchangeable and can be replaced if needed*
 - **StatefulSet**
 - *one or more related Pods that track state*
 - *For example, if a workload records data persistently, run a StatefulSet that matches each Pod with a persistent volume.*
 - **DaemonSet**
 - *Ensures that all (or some) Nodes run a copy of a Pod*
 - *Such as a cluster storage daemon, logs collection, node monitoring running on every node*
 - **Job and CronJob**
 - *Define tasks that run to completion and then stop.*
 - *Jobs represent one-off tasks, whereas CronJobs recur according to a schedule.*

Deployment Spec Example

- Deployment spec

```
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    name: nginx-deployment
5  spec:
6    selector:
7      matchLabels:
8        app: nginx
9    replicas: 3 # tells deployment to run 3 pods matching the template
10   template:
11     metadata:
12       labels:
13         app: nginx
14     spec:
15       containers:
16         - name: nginx
17           image: nginx:1.14.2
18           ports:
19             - containerPort: 80
```

- *A desired state of an application running in the cluster*
- *Kubernetes reads the Deployment spec and starts three app instances*
- *If an instance fails, Kubernetes starts a replacement app instance*

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What is a Service?

- A Kubernetes **Service** is an abstraction that defines
 - *A logical set of Pods*
 - *A policy to access them.*
- Pods are ephemeral – their IPs change when recreated
- A Service provides a stable virtual endpoint for a set of Pods
- Services enable reliable communication between components:
 - *Internal pods communication*
 - *External access to cluster workloads*
- Each Service gets
 - *A DNS name and*
 - *virtual IP (ClusterIP) inside the cluster.*
- Kubernetes component **kube-proxy** manage routing to backend Pods.

Service Types

- **ClusterIP**
 - *Exposes the Service on an internal IP in the cluster only.*
 - *Used for internal communication between Pods.*
- **NodePort**
 - *Exposes the Service on each Node's IP at a static port (e.g. 30080).*
 - *Accessible externally via **NodeIP:NodePort**.*
- **LoadBalancer**
 - *Provisions an external load balancer (e.g. in cloud environments).*
 - *Routes external traffic to the Service.*
- **ExternalName**
 - *Maps the Service to an external DNS name.*
 - *No proxying — pure DNS CNAME redirection.*

How Services Work

- **Selector**

- A Service usually defines a **selector** — a label query used to find matching Pods.
- Example: **selector: app=nginx** matches all Pods with label **app=nginx**.
- Kubernetes monitors Pods that match this selector and updates Service backends

- **Endpoints / EndpointSlice**

- For every Service with a selector, Kubernetes creates an **Endpoints** (or **EndpointSlice**) object listing all healthy Pod IPs and ports.
- This list changes dynamically as Pods are added, removed, or become unhealthy.

- **kube-proxy**

- Runs on every Node and watches Service and Endpoint objects.
- Programs **iptables** or **IPVS** rules to forward traffic from the Service's virtual IP (**ClusterIP**) to one of the backend Pod IPs.
- Load balancing is done using round-robin or IPVS algorithms.

- **DNS Integration**

- **CoreDNS** automatically creates a DNS record for each Service:
 - **<service>.<namespace>.svc.cluster.local**
- Pods can reach the Service via DNS without knowing Pod IPs
 - **curl http://my-service.default.svc.cluster.local**

ClusterIP Service Example

- Example configuration exposing an NGINX Deployment internally:

```
1  apiVersion: v1
2  kind: Service
3  metadata:
4    name: nginx-svc
5  spec:
6    selector:
7      app: nginx
8    ports:
9      - protocol: TCP
10      port: 80
11      targetPort: 8080
12    type: ClusterIP
```

- Pods with `app=nginx` receive traffic through `ClusterIP`.
- DNS name: `nginx-svc.default.svc.cluster.local`
- Used by other Pods to connect via `http://nginx-svc:80`.

Packet Forwarding and Load Balancing

- iptables mode

- *kube-proxy* creates NAT rules in the **nat** table to redirect Service traffic.
- Example traffic coming to **NodeIP:NodePort** (e.g. **192.168.1.11:30080**)

```
1  # 1. Match NodePort traffic coming from outside
2  -A KUBE-NODEPORTS -p tcp --dport 30080 -m addrtype ! --src-type LOCAL \
3    -j KUBE-MARK-MASQ
4    # Mark all external traffic for SNAT (so replies go back via this node)
5
6  # 2. NodePort forwards traffic to the Service chain
7  -A KUBE-NODEPORTS -p tcp --dport 30080 \
8    -m comment --comment "default/my-service: NodePort" \
9    -j KUBE-SVC-XYZ123
10
11 # 3. Service chain chooses one backend Pod
12 -A KUBE-SVC-XYZ123 -m statistic --mode random --probability 0.5 \
13   -j KUBE-SEP-A1B2C3
14 -A KUBE-SVC-XYZ123 -j KUBE-SEP-D4E5F6
15
16 # 4. Pod DNAT rule to redirect to Pod IP:port
17 -A KUBE-SEP-A1B2C3 -p tcp -m tcp -j DNAT --to-destination 10.42.0.12:8080
18 -A KUBE-SEP-D4E5F6 -p tcp -m tcp -j DNAT --to-destination 10.42.1.7:8080
```

- The node's routing table determines how to reach the Pod's IP:
 - **10.42.0.0/24** via **192.168.1.12** dev **flannel.1**
 - packets to Pods in **10.42.0.0/24** (running on Node 2) are sent through the VXLAN interface **flannel.1** to Node 2's IP **192.168.1.12**

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Advanced Topics

- Custom APIs and Controllers
 - *CRDs, Operators, reconciliation loops*
 - *Admission webhooks (mutating/validating)*
- Security
 - *RBAC, Namespaces, Pod Security (seccomp, capabilities, rootless)*
 - *Image signing and supply chain (SBOM, cosign), Secret management (Vault/CSI)*
 - *Policy engines: OPA Gatekeeper, Kyverno*
- Networking
 - *CNI, eBPF (Cilium), NetworkPolicies, Ingress*
 - *Gateway API, Service Mesh (mTLS, traffic shaping)*
- Storage
 - *CSI drivers, snapshots, expansion, topology-aware PVs*
 - *Backup/DR (e.g., Velero), StatefulSet patterns*
- Scaling and Scheduling
 - *HPA/VPA/KEDA (event-driven), Cluster Autoscaler*
 - *Affinity/anti-affinity, taints/tolerations, topology spread*
- Ops and Delivery
 - *GitOps (Argo CD/Flux), progressive delivery (canary, blue/green)*